

Bitcoin-Linked Auto Finance

Towards a Bitcoin Standard Cost of Capital

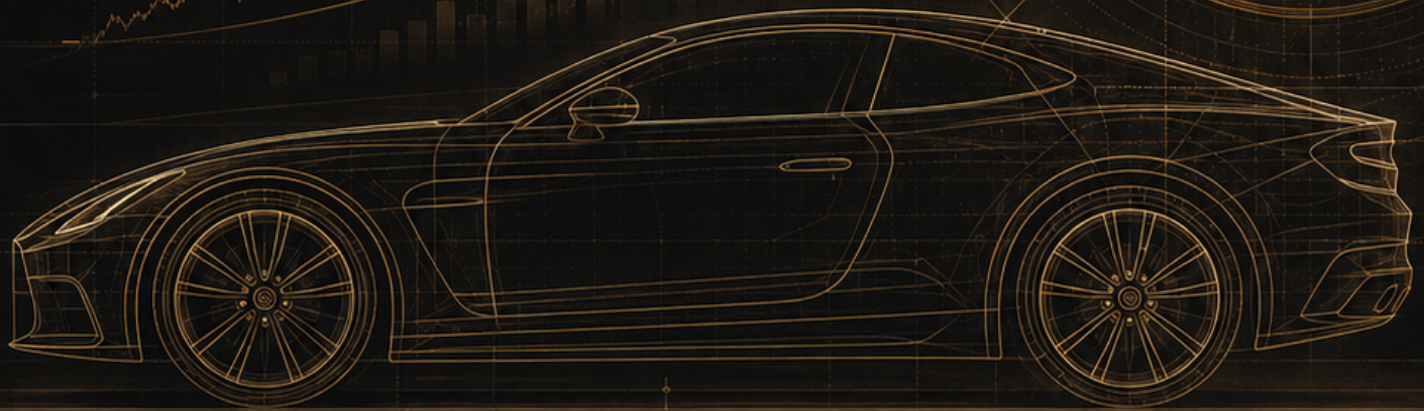
Finance productive assets. Reduce costs. Preserve upside.

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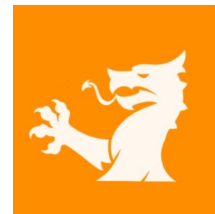
PRODUCTIVE
CREDIT

BITCOIN
STANDARD



Grynvault

Bitcoin-Linked Auto Finance



Towards a Bitcoin Standard Cost of Capital

By Kode / Hudhaifah (aided by AI)

Grynvault exists to finance productive assets and accelerate the world's transition to a Bitcoin Standard. When Bitcoin is embedded into real-world financing, capital can become cheaper for operators, returns can improve for investors, and upside can be preserved instead of surrendered to inflation, interest, and refinancing cycles. Grynvault targets 1% monthly tax-efficient cashflow with downside-protected Bitcoin exposure, beginning with auto finance and expanding toward a broader market for productive assets operating on a Bitcoin standard.

This report uses a historical vehicle finance portfolio as a proxy to test that vision. We backtest and ask a practical question: what would have happened if the customer's original down payment had been allocated into Bitcoin during the life of each contract, and a portion of the upside was used to reduce the customer's final cost while preserving the remainder for investors?

The Cost-of-Capital Problem

Interest is often treated as the price of money. But more precisely, it is a measure of return across time. In business, every investment has an implied rate of return. That rate tells us whether capital was used well, whether the market was beaten, and what could have been done differently.

Discovering a rate after productive activity is different from setting a rate in advance. If someone sells 1 kg of salt for \$5, that is a price. It remains a price whether the salt is delivered today or tomorrow. But if someone exchanges 1 kg of salt today for 1.1 kg of salt tomorrow, that is no longer a simple price. It is a time-based claim on more units of the same asset.

When prices are true, markets can discover value directly. They do not need a central bank rate on salt to decide what delayed delivery should cost. But when money itself is managed through administered interest rates, everyone begins pricing time through the banking system instead of through real productive exchange. At scale, this distorts the price function of markets. It incentivizes credit expansion, fractional reserve banking, monetary inflation, and ever-rising nominal prices despite technological progress making goods and services easier to produce.

Grynvault is built around a different premise: cost of capital should be discovered through productive asset performance, collateral quality, monetary discipline, and final settlement outcomes — not imposed through an arbitrary rate disconnected from the asset being financed.

The New Grynvault Model

Driven to think different, we share here an operating design for financing productive assets while embedding Bitcoin (BTC) into the cost-of-capital architecture from day one. It is an illustrative model, applicable to any asset. Consider that an investor contributes \$100,000, and an operator contributes \$25,000, so a structure between them finances a

\$100,000 car. After a \$6,250 fee we might keep, \$18,750 is allocated to a Bitcoin-linked sidecar or reserve, and a predictable monthly gross repayment of approximately \$2,125 begins.

This targets an investor IRR of approximately 8.00%, and an operator cost of capital of approximately 15.71% assuming BTC is flat and boring. Now here are the twists: if BTC is up, costs drop and returns rise; if BTC is down, costs rise and returns stay put. The cashflow and embedded variable rate that emerges from this product is more grounded and presents a more asymmetric approach to Bitcoin business. Compared to a pure Bitcoin overlay, investors gain convexity and optionality while operators gain cost advantages and liquidity. The initial contributions from the repurposed down-payment allows the operator to make stable payments, and for the Bitcoin-linked volatility to hide in the final result instead of disturbing monthly affordability.

The backtest detailed later is therefore best understood as a proxy. It does not perfectly replicate the newer Grynvault structure, but it tests the same economic question: can productive asset financing become more efficient when part of the deal is linked to an appreciating monetary asset? The answer, historically, appears to be yes.

Model item	Illustrative value
Investor contribution	\$100,000
Operator contribution	\$25,000
Car acquisition	\$100,000
BTC-linked sidecar / reserve	\$18,750
Upfront fee	\$6,250
Monthly gross repayment	\$2,125
Investor IRR	8.00%
Operator cost of capital	15.71%

Source: Final Grynvault Subprime Auto model, 5ybtc (subprime-Car) tab.

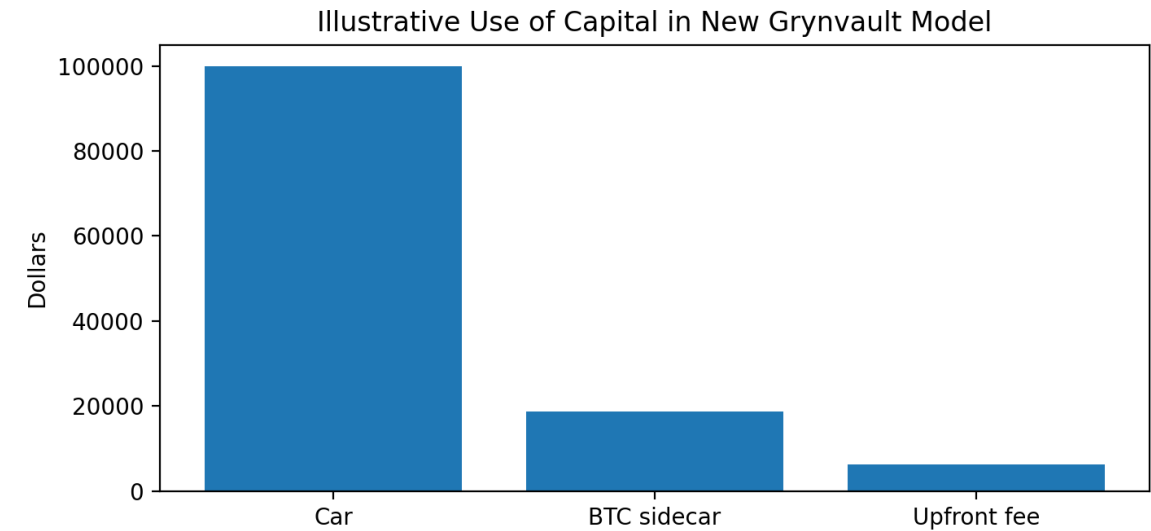


Figure 1. The newer Grynvault model embeds Bitcoin as part of the deal architecture rather than as a separate speculative trade.

The Model in Plain English

Traditional financing treats the borrower's cost and the investor's return as opposite sides of the same trade. If the investor earns more, the borrower usually pays more. If the borrower pays less, the investor usually earns less. Grynvault is exploring a different possibility to align incentives almost like equity. Instead of relying only on interest, refinancing, or yield-curve spreads, the model embeds Bitcoin into the economics of each deal. The productive asset still does the real-world work. The car, truck, equipment, or business asset still generates utility and cashflow. But the monetary layer is no longer passive. It participates.

In the historical backtest, the customer's down payment became the Bitcoin-linked component. That Bitcoin exposure was held through the contract period. At the end, part of the gain was used to reduce the customer's final buyout or create money back, while the remaining gain improved investor returns. In the live Grynvault model, this same logic is built into the structure from the start, with clearer downside management, cashflow discipline, operator performance rules, and final settlement mechanics.

The Backtest

Using historical lease data, each contract was reconstructed and matched with Bitcoin prices from the contract start date to the relevant end or settlement date. Everything about the original financing stayed the same: the vehicle, the payment schedule, defaults, losses, underwriting record, and operational performance. The only change was the monetary layer.

At the end of each contract, the model tested two scenarios. In the first, the customer received 30% of the Bitcoin gain above the original down-payment value. In the second, the customer received 50%. The rest of the gain remained available to the investor. The point was not to show that Bitcoin can replace the lease. The point was to see whether the same real-world asset cashflows could produce better outcomes if paired with an appreciating monetary asset.

What Happened

Across the full modelled portfolio, the original average IRR was positive. The active book was much stronger than the terminated book, while terminated and default-heavy paper showed the drag of real-world credit friction. This is exactly why the backtest is useful: it is not a clean marketing sample. It contains good deals, weak deals, active deals, early exits, defaults, and operational noise. The table below separates the portfolio into the right lenses.

Portfolio lens	Simple avg IRR	Weighted IRR	Interpretation
Full valid modelled portfolio (138 deals)	8.05%	6.87%	Best headline base case for modeled portfolio economics
Active book (62 deals)	16.85%	16.62%	Shows scheduled economics of contracts still running
Terminated book (76 deals)	0.87%	-2.73%	Shows drag from defaults, early exits, insurance, and other realized friction
Observed stress lens (86 deals)	2.65%	-0.38%	Useful as a stress view, not a portfolio headline

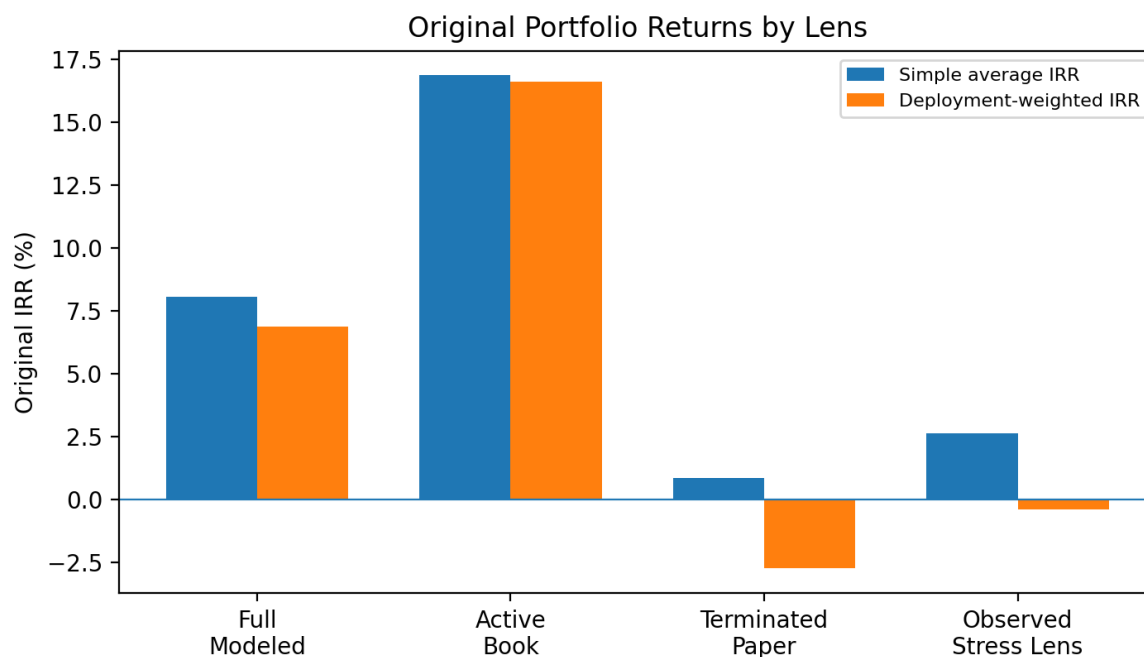


Figure 2. Active deals materially improve the base return picture and should not be excluded from the portfolio view.

When the Bitcoin sidecar is applied to the full modelled portfolio, the investor picture improves materially. The original full portfolio average IRR of 8.05% increased to 25.75% in the 30% borrower-share scenario and 24.10% in the 50% borrower-share scenario. On a deployment-weighted basis, the original 6.87% base improved to 22.73% and 21.25%, respectively.

The borrower side also improved. Average borrower benefit was approximately \$1,884 in the 30% model and \$3,140 in the 50% model, with median benefits of approximately \$1,119 and \$1,866. In the APR-comparable observed subset from the deeper reconstruction, borrower effective APR fell from 16.18% to 9.53% under the 30% share and to 6.55% under the 50% share.

Scenario	Simple avg investor IRR	Weighted investor IRR	Improvement vs avg base
Original portfolio	8.05%	6.87%	-
30% borrower BTC share	25.75%	22.73%	+1770 bps avg
50% borrower BTC share	24.10%	21.25%	+1605 bps avg

Table 2. Full modeled investor outcomes under the BTC-linked sidecar.

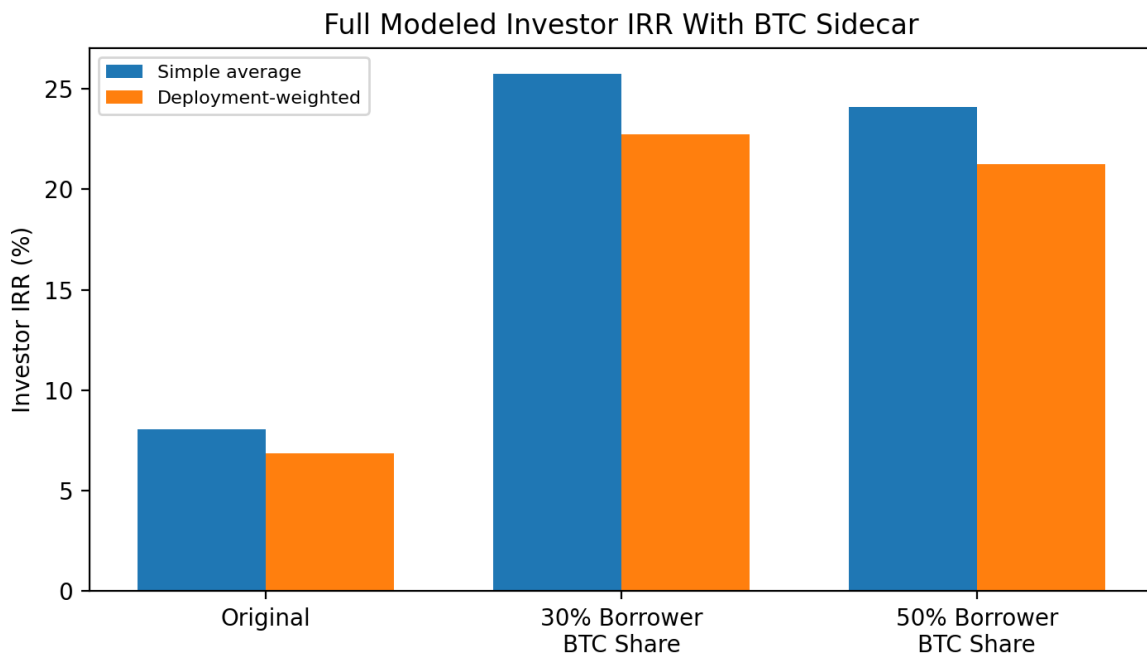


Figure 3. The BTC sidecar improves investor IRR in the full modeled portfolio while still sharing upside with borrowers.

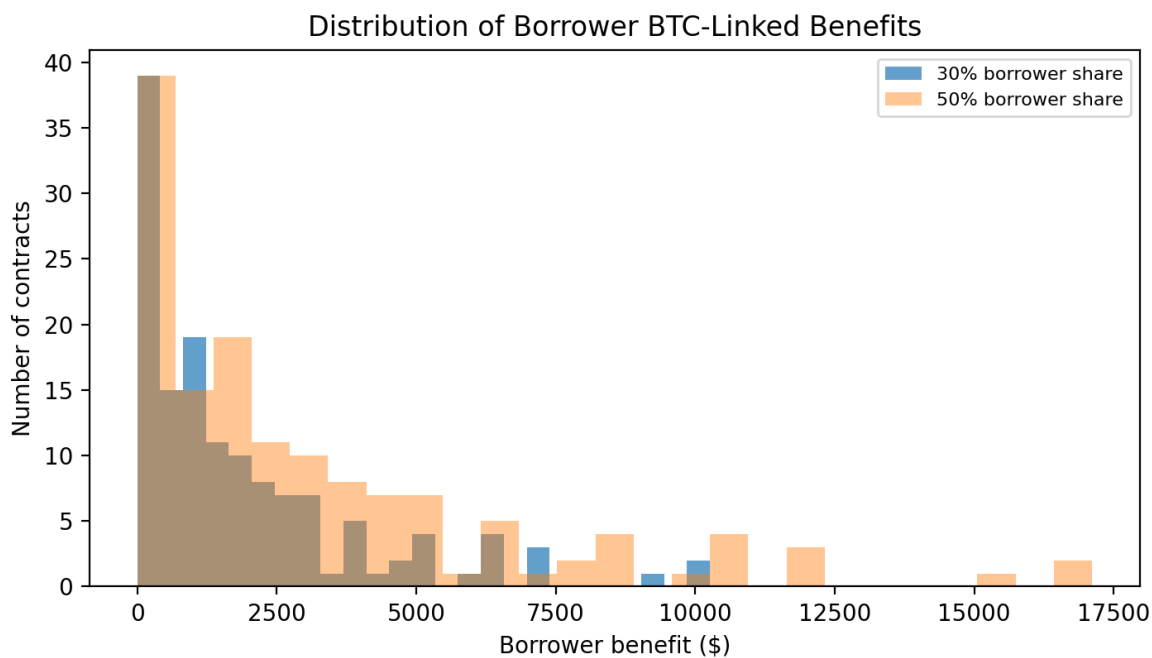


Figure 4. Borrower benefits are positively skewed, which is expected when the linked monetary asset has convex upside.

Why Auto Finance

Auto finance is not the final vision. It is the wedge. Cars are useful because they expose the problem clearly: they are real assets, they amortize quickly, they have observable market values, and they create a familiar financing experience. They also reveal how expensive capital can become for ordinary borrowers and operators.

Auto finance is a practical testing ground for the Bitcoin standard cost of capital. If Bitcoin can reduce financing friction in a market as operationally grounded as vehicles, then the same logic can potentially extend to equipment, inventory, trucks, machines, real estate-adjacent assets, and operating businesses. The long-term idea is not Bitcoin car loans. The long-term idea is productive finance on a Bitcoin standard.

Why This Matters

Most financing systems are built around inflationary money. They assume debt expansion, refinancing risk, rate cycles, and purchasing-power erosion as normal features of the system. A Bitcoin standard changes the premise. If the monetary asset embedded inside the deal appreciates over time, then financing no longer needs to depend only on extracting more from the borrower. Part of the return can come from preserving and sharing monetary upside. That changes the emotional and economic relationship between borrower and investor.

The borrower is no longer simply paying for access to capital. The borrower is participating in a structure where good performance can reduce the final cost. The investor is no longer relying only on fixed interest or spread. The investor participates in productive cashflow and Bitcoin-linked upside. This is how financing begins to feel less extractive and more aligned.

Addressing Downsides

The strongest criticism of this concept is obvious: it only works if Bitcoin goes up. That criticism is partially fair. Bitcoin is volatile, and a structure that blindly assumes perpetual appreciation would be fragile. But that is not what Grynvault is building.

Finance is about structuring risk. The important question is not: does Bitcoin go up, but rather, how is downside managed?

In the Grynvault model, Bitcoin volatility is included as a cost to the operator, almost like a variable rate connected to BTC price movement, but implemented smoothly with no impact on monthly payments and only changing the final result. This means that even if Bitcoin went to zero, the worst case outcome would just be the regular stress-lensed result, with maximum cost of capital. This is roughly capped at an effective rate of 24%. High, but not uncommon in the subprime field.

The backtest should not be read as a promise of future BTC appreciation. It should be read as evidence that the historical monetary tailwind was large enough to change deal economics, and that a structured version of the concept can allocate that tailwind through rules, reserves, performance gates, and final settlement mechanics rather than through unmanaged speculation.

Conclusion

Traditional finance says: pay interest for access to capital. We ask a different question: what if productive assets could help people move toward a Bitcoin standard instead? The next step is to build the structure carefully enough that when Bitcoin does what it was designed to do, productive businesses, operators, and investors are positioned to benefit together. That is the foundation for a Bitcoin standard cost of capital.